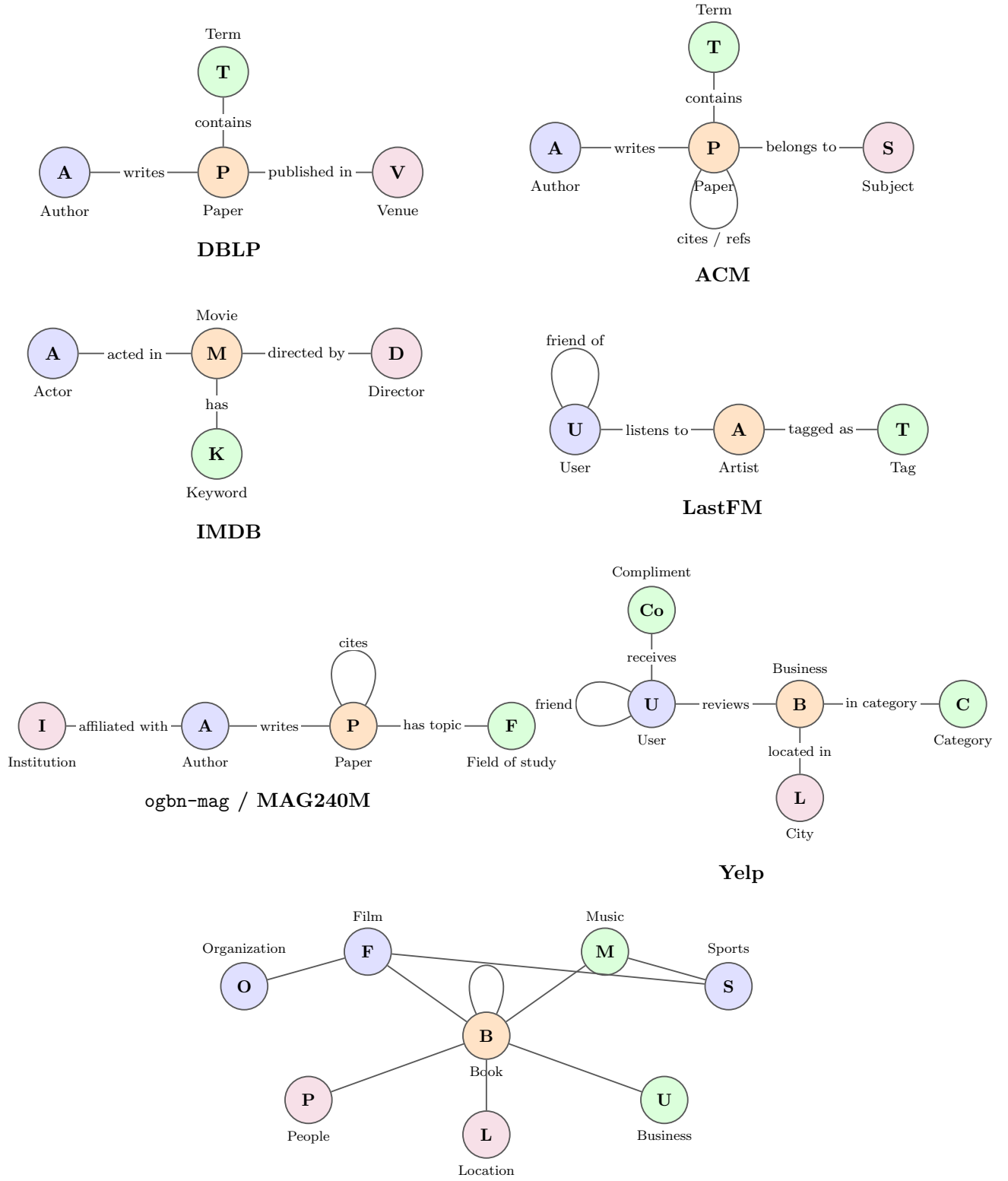


3. Network schemas and meta-paths

Each schema below is $T_G = (\mathcal{A}, \mathcal{R})$ for one benchmark family: circles are node types (letter = the initial used in meta-path notation), edges are relation types. Inverse relations are omitted for readability.

Table 1:

Dataset	Meta-paths	Semantics
DBLP	APA	two authors co-wrote a paper (co-authorship)
	APCPA / APVPA	two authors published at the same venue (community)
	APTPA	two authors used the same term (topical similarity)
ACM	PAP	two papers share an author
	PSP	two papers share a subject
	PcPAP, PrPSP	HGB variants routed through a citation (c) or reference (r) hop
IMDB	MAM	two movies share an actor
	MDM	two movies share a director
	AMA, DMD	two actors co-starred / two directors share a movie
ogbn-mag	PAP	two papers share an author
	PAIAP	two papers from the same institution (via author-institution)
LastFM	UAU	two users listen to the same artist
	UATAU	two users like artists sharing a tag
	AUA, ATA	two artists share a listener / a tag
Yelp	UBU	two users reviewed the same business
	UBCaBU	two users reviewed businesses of the same category
Freebase	BB, BFB, BPB	book-book; books sharing a film / a person



Freebase (HGB) — 8 types, 36 relations; a representative subset is drawn.

Figure 1: